Live Feed-Based PPE Monitoring and Safety Compliance System

A project concept, architecture, evaluation plan, and responsible-use overview

Final-Year Project Background Document

This document describes a proposed system design for academic discussion and summarization testing. It does not claim that the system, model, dataset, or workplace deployment has already been implemented or validated.

Approximately 20,088 characters of selectable text

1. Abstract

Live Feed-Based PPE Monitoring and Safety Compliance System is a proposed computer vision application for helping safety teams observe whether required personal protective equipment is being worn in designated work areas. The system receives video from an authorized live camera feed, analyzes sampled frames, detects people and selected protective equipment, associates equipment observations with people where possible, and presents evidence for human review. Its intended role is decision support. It should not make disciplinary, employment, or emergency decisions on its own. The design emphasizes reliable observation, privacy, transparent limitations, and a clear path from detection to a corrective safety action. This document describes a project concept and a practical implementation plan. It does not claim that a particular model, camera installation, dataset, accuracy level, or deployment has already been completed.

2. Introduction

Industrial and construction workplaces often require workers to use equipment such as hard hats, high-visibility vests, protective eyewear, gloves, and safety footwear. These requirements depend on the hazards in each area and on the organization's approved safety procedures. A missed item can increase the chance of injury, while a false alarm can distract workers and safety staff. Manual observation remains essential, but supervisors cannot continuously inspect every camera view and every worker. A computer vision tool may help direct attention to moments that deserve a closer look. The project explores how a live video feed can be processed to identify visible PPE indicators and communicate possible non-compliance in a timely, reviewable way.

3. Problem statement

The central problem is the difficulty of consistently identifying visible PPE compliance across one or more monitored work zones. Existing camera footage is often used only after an incident, while safety teams may need earlier indications that an item appears to be missing. A useful system must cope with movement, changing light, partial occlusion, different camera angles, crowded scenes, and equipment that varies in color or design. It must also distinguish an actual absence from an item that is hidden or too small to judge. The proposed system therefore treats each automated result as a candidate observation with a confidence value and supporting frame, rather than as unquestionable proof of a violation.

4. Aim and objectives

The aim is to design and evaluate a live feed-based PPE monitoring prototype that can detect selected visible safety equipment and present possible compliance events to a human reviewer. The project objectives are: (1) define a limited set of PPE categories and work-zone rules; (2) obtain or prepare appropriately licensed, representative image data; (3) train or adapt a suitable object-detection model; (4) process a live or recorded video source at a practical sampling rate; (5) associate detections with people when the view supports that association; (6) present time-stamped alerts and evidence in a review interface; (7) measure detection and event-level performance on held-out data; and (8) document privacy, safety, and deployment limitations. Objectives should be refined to match available time, hardware, data, and institutional requirements.

5. Scope and assumptions

A manageable first version should monitor one camera or a recorded test feed and a small set of clearly visible PPE classes, for example hard hats and high-visibility vests. The system should not claim to identify every kind of PPE or infer whether equipment is correctly fitted unless the model and evaluation specifically support that claim. It should not identify workers by name or use facial recognition. The initial prototype can process frames locally, retain only short-lived event evidence, and allow an authorized reviewer to confirm or dismiss an alert. Real-time performance, multiple-camera scaling, and integration with access-control or human-resources systems are outside the initial scope unless they become explicit, ethically reviewed requirements.

6. Proposed system overview

The proposed pipeline begins with a camera or video source configured by an administrator. A capture component reads frames and assigns timestamps. A sampling component avoids needlessly analyzing every frame, while preserving enough temporal information to notice a person entering or leaving the monitored area. An object detector proposes bounding boxes and labels for people and PPE. A matching stage estimates which equipment belongs to which person, using spatial relationships and, if justified, a lightweight tracker across nearby frames. A rule engine applies the PPE requirements for the configured zone. A temporal filter groups repeated observations into an event rather than creating a new alert for every frame. Finally, a dashboard displays the event, camera, time, detected categories, confidence, and a limited evidence image so a reviewer can make a decision.

7. Architecture and components

Video source: an authorized RTSP stream, USB camera, or recorded sample video. **Frame capture:** OpenCV or an equivalent library reads frames, records timestamps, and reports connection failures. **Inference service:** a trained detector runs on sampled frames using CPU or an available GPU. **Association and temporal logic:** detections are linked conservatively to person boxes and grouped over time. **Policy configuration:** each camera is mapped to a zone and a list of required visible PPE classes. **Event store:** a database records minimal metadata and review state; evidence images have a defined retention period. **User interface:** authorized safety personnel review alerts and mark them confirmed, dismissed, or needing follow-up. **Monitoring:** logs capture system health and errors without unnecessarily storing video or personal data.

8. Detection and person-equipment association

An object detector returns candidate objects and confidence scores for each frame. A person bounding box can be compared with PPE boxes using overlap, relative position, and scale. For example, a hard-hat candidate near the top of a person box may be associated with that person when both detections are sufficiently clear. This geometric approach is imperfect: a nearby person's helmet can overlap another person's box, and a worker bending down may violate simple position assumptions. Tracking IDs can help maintain continuity for a short interval, but they are not identity labels and should not be treated as proof that the same individual is present across long periods. Ambiguous associations should be marked unknown and sent for review instead of being converted into confident violations.

9. Compliance rules and event generation

A zone rule specifies which categories are required and, if appropriate, which areas or tasks the rule applies to. A missing-PPE candidate should be generated only when a person is sufficiently visible, the relevant body region can reasonably be assessed, and the detector has had adequate opportunity to observe the scene. The system can require repeated evidence across several sampled frames before creating an event. Hysteresis or a cooldown period can reduce rapid alert toggling when a detection confidence fluctuates near a threshold. Each event should carry the rule applied, the evidence timestamps, confidence information, and uncertainty flags. The user interface should distinguish an automated candidate from a reviewer-confirmed incident. A worker leaving the camera view should not automatically be interpreted as a safety violation.

10. Dataset preparation

Training data should represent the intended cameras and operating conditions as closely as possible. Images may come from a suitably licensed public PPE dataset, staged recordings with informed participants, or an approved internal source. Each image needs consistent annotations for the selected categories, including people, relevant PPE, and difficult or ambiguous examples. Annotation guidance should explain how to label partially visible objects, reflections, occlusion, empty scenes, and uncertain cases. Data should be divided into training, validation, and test sets so that near-duplicate frames or adjacent video frames do not leak across the split. When possible, split by recording session, camera, or location to test generalization rather than memorization of a scene. Document class counts, exclusions, licensing, and any class imbalance.

11. Model development

A pretrained object detector can provide a practical starting point, with fine-tuning on the chosen PPE classes if the dataset and compute budget permit. The project should record the model family, pretrained checkpoint, software versions, image resolution, training parameters, and random seed. Baselines are useful: a simple pretrained detector or a conservative confidence threshold can show whether a more complex approach adds value. Training and validation decisions must not use the final test set. Data augmentation may help with scale, brightness, and viewpoint variation, but should not create unrealistic examples that obscure the safety task. The model should be versioned together with its class mapping and rule configuration so results can be reproduced. A model card or project note should describe intended use and known failure cases.

12. Live video processing

Live capture must handle latency, dropped frames, camera disconnection, and changes in stream resolution. The processing loop should avoid an unbounded queue: when inference is slower than capture, it is often safer to skip stale frames and analyze a recent one than to build a backlog that produces outdated alerts. The system should expose capture status, last-frame time, inference duration, and alert-processing delay. Frame sampling can be configured based on measured performance, but its effect on missed short-lived events must be evaluated. A local recorded-video mode is valuable for repeatable testing and demonstrations. It allows an operator to compare outputs across model versions without connecting to a real workplace feed.

13. User interface and workflow

The dashboard should show which camera or test video is active and whether the feed is healthy. An alert card should include the time, zone, PPE category believed to be missing, confidence, and a compact evidence frame. Reviewers should be able to confirm, dismiss, or mark an event for further inspection, with a short reason when appropriate. The interface should make clear that alerts are machine-generated candidates. A summary view can show counts by camera, category, and time period, but counts must distinguish unreviewed candidates from confirmed events. Controls should support pausing processing and managing evidence retention. Keyboard access, readable status labels, and clear error messages are necessary because operators may use the system under time pressure.

14. Evaluation methodology

Evaluation should be performed on data that was not used for training or threshold selection. At the object level, report precision, recall, and average precision for each PPE category, along with a confusion matrix where appropriate. Precision describes how many predicted detections were correct; recall describes how many labelled objects were found. Neither alone is sufficient for a safety workflow. At the event level, measure whether a true non-compliance episode generated an alert, how many false alerts occurred per hour of footage, and the delay between visible evidence and alert creation. Report results by lighting, distance, occlusion, camera angle, and other relevant conditions when sample sizes permit. Include confidence intervals or counts so small evaluations are not overstated. A human review study can assess whether evidence is understandable and whether alerts help reviewers.

15. Risks, limitations, and failure cases

Common failure modes include an item being hidden by another person, a hard hat blending into the background, reflective clothing under unusual lighting, a camera pointing away from the work area, motion blur, low resolution, and people appearing only partially in frame. A model may confuse similar objects or perform worse for equipment styles and environments absent from training data. False negatives can create a misleading impression that the area is compliant; false positives can produce alert fatigue and unfairly burden workers. The detector cannot determine whether PPE is certified, undamaged, properly worn, or appropriate to a particular hazard unless those attributes are observable and specifically validated. A high object-detection score does not establish that the full safety process is effective. Human observation and established safety controls remain necessary.

16. Privacy, fairness, and responsible use

Video monitoring can affect worker privacy and trust. Before any real deployment, the organization should identify a lawful and legitimate purpose, consult relevant stakeholders, provide appropriate notice, restrict access, and follow applicable rules and institutional review requirements. Prefer processing locally and retaining event metadata rather than continuous footage. If evidence images are needed, minimize their resolution and retention period while preserving enough context for a fair review. Do not add face recognition or use the system to infer identity, productivity, emotion, or unrelated behavior. Evaluate performance across relevant clothing, body types, skin tones, lighting conditions, and work roles where those factors influence visual results, while avoiding unsupported demographic inferences. Provide a way to challenge or correct an alert. Automated results should not be the sole basis for disciplinary action.

17. Security and operational safeguards

Camera credentials and stream URLs should not be hardcoded in source code or exposed in logs. Access to feeds, event records, and evidence should follow least-privilege principles. Use encrypted network connections where supported and protect stored credentials with an appropriate secret-management method. The application should log operational events such as camera disconnects, model errors, and reviewer actions, but logs should avoid unnecessary personal data. Define backup, retention, deletion, and incident-response procedures before deployment. A fail-safe design should show a visible unavailable state when a feed or inference service stops; it must not silently represent missing observations as compliance. Changes to the model, thresholds, cameras, or PPE policy should be recorded and reviewed.

18. Information technology and systems concepts

This project connects computer vision with information systems engineering. It has an input layer for video, a processing layer for detection and rule evaluation, a persistence layer for minimal event records, and a presentation layer for review. Requirements analysis defines what evidence is needed and how users act on it. Data engineering addresses annotation, data quality, and leakage. Software engineering contributes modular components, error handling, configuration, and reproducible tests. Evaluation links model behavior to operational outcomes instead of reporting only a demonstration screenshot. Human-computer interaction determines whether alerts are understandable and actionable. The system should be assessed as a sociotechnical workflow: a detector alone cannot ensure compliance, and a well-designed review process cannot compensate for a camera that fails to capture the required evidence.

19. Testing strategy

Unit tests can validate confidence filtering, box association, temporal grouping, zone rules, event state changes, and evidence-retention decisions. Integration tests can feed a short recorded clip through capture, inference, and alert storage. A mock detector makes it possible to test the event workflow without downloading or running a large model. Dataset tests should check that annotations fall within image bounds, class IDs are valid, and split membership has no accidental overlap. Reliability tests should simulate a camera disconnect, corrupt frame, slow inference, empty detection result, and database outage. Performance checks should measure frames processed per second, inference latency, memory use, and alert delay on the intended hardware. A demonstration should include both successful detections and known failure examples, not only ideal scenes.

20. Implementation roadmap

A staged implementation reduces risk. **Stage 1:** confirm project requirements, safety rules, stakeholder expectations, and ethical constraints. **Stage 2:** choose a small PPE class set and obtain a legal, representative dataset; document annotations and split strategy. **Stage 3:** establish an image-based detection baseline and evaluate it on held-out examples. **Stage 4:** add recorded-video frame capture, timestamps, and detection visualization. **Stage 5:** implement cautious person-equipment association and event grouping. **Stage 6:** add a review dashboard and minimal event storage. **Stage 7:** test operational failures, evaluate event-level behavior, and improve known weaknesses. **Stage 8:** prepare documentation, a reproducible demonstration, and a clear account of what remains future work. Each stage should have acceptance criteria agreed with the academic supervisor.

21. Expected outcomes

A successful academic prototype would demonstrate an end-to-end path from a camera or recorded video to a human-reviewable PPE candidate event. It would include a documented set of supported categories, a trained or adapted detector, a repeatable test split, object-level and event-level metrics, and a user interface that shows evidence and uncertainty. The expected outcome is not a guarantee of perfect compliance detection. The project should report measured strengths and weaknesses, including conditions under which it should not be relied upon. A useful contribution may be a well-scoped, transparent workflow and a rigorous evaluation, even if measured accuracy is limited. Claims in the final report should correspond directly to completed implementation and experiments.

22. Conclusion

A live feed-based PPE monitoring system can help safety personnel direct attention toward possible missing equipment, but it must be designed as a cautious support tool. Reliable use depends on representative data, robust camera operation, conservative event logic, meaningful evidence, fair review, privacy safeguards, and evaluation that reflects real operating conditions. The proposed project combines object detection, video processing, event management, and human-centered system design. Its practical value should be judged not only by how well a model recognizes objects, but also by whether alerts are timely, understandable, and appropriately acted upon. Any real workplace use requires organizational approval, applicable legal review, worker engagement, and continued human oversight.

Appendix: Suggested project report evidence

For a clear final-year demonstration, retain a requirements table, system architecture diagram, data-source and license record, annotation guide, train-validation-test split description, model configuration, and software environment. Include representative examples of true detections, missed detections, false detections, occlusions, and camera failure. Present the number of test images or video events behind each metric. Separate prototype features from planned features. Explain how a reviewer can correct an alert and how long evidence is retained. Keep a short demonstration video or recorded test clip that can be replayed consistently. Do not publish private camera footage, worker images, credentials, or internal safety documents in a public repository. A reproducible report is more informative than an unsupported claim that the system operates in real time or achieves high accuracy.